

Forward 3-point difference $[O(\omega^2)]$:

$$\delta'_i = \frac{-3\delta_i + 4\delta_{i+1} - \delta_{i+2}}{2\Delta\omega}$$

Backward 3-point difference $[O(\omega^2)]$:

$$\delta'_i = \frac{\delta_{i-2} - 4\delta_{i-1} + 3\delta_i}{2\Delta\omega}$$

Central 5-point difference $[O(\omega^4)]$:

$$\delta'_i = \frac{\delta_{i-2} - 8\delta_{i-1} + 8\delta_{i+1} - \delta_{i+2}}{12\Delta\omega}$$